Grey-box Model of the Profondeville Energy Center for Demand-Response MPC

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Abstract

This note derives a control-oriented grey-box model of the Profondeville district heating energy center, for the purpose of a demand-response (DR) model predictive controller (MPC) driven by electricity price, PV production and a district consumption forecast. The physical boundary of the model is the *supply temperature to the district*; the apartments are not modelled. The controllable subsystem consists of the air-source heat pump (ASHP, modulating), the geothermal heat pump (GSHP, on/off) and the distribution 3-way mixing valve. We present (i) a detailed plant model suitable for system identification, keeping the two internal PID loops explicit, and (ii) a reduced model suitable for online optimisation, in which the PID loops collapse into algebraic set-point tracking plus feasibility constraints. A single continuous input (the ASHP) and a single storage state (the buffer tank) are sufficient for a first controller; the geothermal on/off unit and a stratification (thermocline) state are documented as upgrades.

Contents

1 System and modelling scope

In winter the energy center produces hot water into a $2.5\,\text{m}^3$ buffer tank, charged by two heat pumps, and feeds a district network whose supply temperature is regulated by a 3-way mixing valve. The relevant energy flows are:

- **ASHP** (air source, $\sim$20 kW): inverter-*modulating*, 0–100 %. Its onboard controller regulates the leaving-water temperature to a set-point. This is the primary continuous actuator.
- **GSHP** (ground source, $\sim$40 kW): *on/off* only. Treated in v1 either as off or as a measured exogenous heat input.
- **Buffer tank** ($2.5\,\text{m}^3$): the thermal storage that provides the demand-response flexibility.
- **3-way valve** (`FCV_702`): blends hot tank water with the warm district return to hold the district supply temperature on a weather-compensated heating curve.

Measurement reality (design constraint). Water *flows* and *heat meters* on the heat-pump loops are *not* currently instrumented and are assumed unavailable. Only temperatures, valve/pump positions, the district flow `FQT_701_OUT`, and heat-pump *electrical* power are measured. The model is therefore built so that all unmeasured flows are absorbed into identifiable parameters, and the heat-pump heat input is inferred from measured electrical power.

Control architecture (cascade). The MPC does not command power directly. It commands a *set-point*; the electrical power and the delivered heat are *responses* of the heat pump:

$$\underbrace{u = T_{\text{sp}}\ (+\,\text{enable}\ \delta)}_{\text{MPC decision}} \xrightarrow{\text{HP map}} (P_{\text{el}},\ Q_{\text{hp}}) \xrightarrow{\text{tank}} (T_{\text{top}}, T_{\text{bot}}) \xrightarrow{\text{valve}} T_{s,d}. \tag{1}$$

2 Nomenclature and measured signals

Constants: c = specific heat of water (kJ/(kg K)), ρ = density. Flows are expressed as mass flows $\dot{m} = \rho\dot{V}$.

3 Heat-pump model (ASHP)

3.1 Internal set-point-tracking loop (PID → modulation)

The inverter modulates the compressor, $m \in [0, 1]$, to drive the leaving-water temperature T_{hs} to its set-point T_{sp}. This is the first of the two internal PID loops:

$$e(t) = T_{\text{sp}}(t) - T_{hs}(t), \tag{2}$$

$$m(t) = \text{sat}_{[0,1]}\Big(K_p\, e(t) + K_i \int_0^t e(\tau)\,\mathrm{d}\tau\Big). \tag{3}$$

The gains K_p, K_i are ***not identified****. Equation (3) is documentation of the onboard controller, not a block that is fitted. The data already contains this loop's* output*: the modulation, P_{el} and T_{hs} are all logged. For identification (§8) we regress on those measured signals; for control (§7) we collapse the loop to $T_{hs} = T_{\text{sp}}$. The gains enter neither.*

Symbol	Meaning	Unit	BigQuery column
States			
T_{top}	Tank top temperature	°C	`TT_601`
T_{bot}	Tank bottom temperature	°C	`TT_602`
T_{hs}	ASHP supply (condenser outlet)	°C	`PAC_501_T_OUT_EAU_ECHANGEUR`
T_{hr}	ASHP return (condenser inlet)	°C	`PAC_501_T_IN_EAU_ECHANGEUR`
V_{top}	Hot-zone volume (optional, thermocline)	m^3	— (observer)
Control inputs			
T_{sp}	ASHP leaving-water set-point	°C	`PAC_501_CONSIGNE_ACTUEL`
δ	ASHP enable (on/off)	—	`PAC_501_RUN`/`_LIBERATION`
θ	3-way valve position	—	`FCV_702_POSITION`
δ_g	GSHP enable (v1.5)	—	`PAC_301_RUN`
Measured disturbances / forecasts			
T_{air}	Outdoor air temperature	°C	`TT_Text`
Q_{dist}	District heat load (forecast; tank driver)	kW	forecast / $\dot{m}_d c \Delta T$
$\dot{m}_d$	District draw flow (historical only)	m^3/h	`FQT_701_OUT`
$T_{\text{ret},d}$	District return temperature	°C	`TT_702`
T_{amb}	Plant-room ambient temperature	°C	(assumed / const)
$\lambda(t)$	Electricity price	EUR/kWh	external
$\text{PV}(t)$	PV production / forecast	kW	external
Measured outputs			
P_{el}	ASHP electrical power	kW	`PAC_501_PUISSANCE` / `EM_PAC_501_PWR_TOT_P`
$T_{s,d}$	District supply temperature	°C	`TT_701`

Table 1: Model variables and the logged signals they map to. All parameters ($C_\bullet, \kappa, UA_\bullet, \alpha, \beta, \dots$) are identified; unmeasured flows are folded into them.

The modulation delivers thermal power to the condenser water, capped by the air-temperature-dependent capacity ceiling:

$$Q_{\text{hp}}(t) = m(t)\,\dot{Q}_{\max}\big(T_{\text{air}}(t)\big), \qquad \dot{Q}_{\max}(T_{\text{air}}) \approx a_0 + a_1\,T_{\text{air}}. \tag{4}$$

3.2 Electrical power and COP

The electrical power (the quantity multiplied by price) follows from the coefficient of performance, which decreases with lift (sink temperature T_{hs} up, source temperature T_{air} down):

$$P_{\text{el}}(t) = \frac{Q_{\text{hp}}(t)}{\text{COP}\big(T_{\text{air}}, T_{hs}\big)}. \tag{5}$$

Two identifiable COP forms:

$$\text{(Carnot-scaled)} \quad \text{COP} = \eta\,\frac{T_{hs} + 273.15}{T_{hs} - T_{\text{air}} + \Delta T_{\text{pp}}}, \qquad \text{(affine)} \quad \text{COP} = b_0 + b_1 T_{\text{air}} + b_2 T_{hs}. \tag{6}$$

Since $P_{\text{el}}, T_{\text{air}}, T_{hs}$ are all measured, (5)–(6) are fit by regression on historical data. This is the only place nonlinearity enters the cost; Section 7 discusses convex handling.

3.3 Condenser-loop dynamics (optional refinement)

The heat-pump water loop draws from the tank bottom, is heated in the condenser and returns to the tank top. The condenser circulator is *fixed-speed*, so the loop flow $\dot{m}_h$ is a **single unknown**

constant, not a signal. With loop thermal capacitance C_h:

$$C_h \dot{T_{hs}} = Q_{\text{hp}} - \dot{m}_h c (T_{hs} - T_{hr}), \tag{7}$$

$$\tau_r \dot{T_{hr}} = T_{\text{bot}} - T_{hr}. \tag{8}$$

Equation (8) is a first-order lag capturing pipe/mixing transport from the tank bottom to the condenser inlet; in the fast limit $T_{hr} \to T_{\text{bot}}$.

> *On the unknown flow $\dot{m}_h$. It is never needed in physical units. (i) In* (7) *it only appears as the ratio $\dot{m}_h c/C_h$, one lumped coefficient identified from the T_{hs} transient. (ii) For the* heat delivered to the tank *we do not use* (7) *at all: we take it from the measured electrical power, $Q_{\text{hp}} = \alpha P_{\text{el}}$ (§5). Consequently the tank model below contains no $\dot{m}_h$: the flow's only remaining footprint is the recirculation* stirring *it causes, which lumps into a single mixing coefficient κ. This condenser-loop block is therefore an* optional *refinement; the reduced model drops it entirely.*

4 Geothermal heat pump (GSHP)

The GSHP (∼40 kW) is **on/off only** and, in the present operation, its start/stop is decided by the existing plant logic — it is *not* an MPC decision variable in v1. It must nevertheless be modelled, for one reason: to fit the tank on data segments where the GSHP was running, its heat input has to be accounted for, otherwise the identifier attributes the GSHP's charging to the ASHP or to the tank parameters and the fit is corrupted.

Because the ground-source temperature is nearly constant across a heating season, the on-state heat output is essentially constant. The model is therefore a switched constant source, or — preferred, since it is faithful and needs no COP assumption — the *measured* electrical power scaled by an identified proxy:

$$Q_{\text{gshp}}(t) = \begin{cases} \alpha_g P_{\text{el},g}(t) & \text{(measured-power form, preferred)}, \\ \delta_g(t)\, \bar{Q}_{\text{gshp}} & \text{(switched-constant form)}, \end{cases} \tag{9}$$

with $\delta_g \in \{0, 1\}$ the on/off state (`PAC_301_RUN`), $P_{\text{el},g}$ its electrical power (`EM_PAC_301_PWR_TOT_P` / `PAC_301_PUISSANCE`), and α_g or $\bar{Q}_{\text{gshp}}$ identified. This term enters the tank top node in (11) exactly like the ASHP heat. Structurally the two heat pumps are identical inputs to the tank; the difference is only that T_{sp} (ASHP) is a decision variable whereas δ_g (GSHP) is, for now, an exogenous *measured disturbance*. If the GSHP is later added to the controller, (9) is already the required model (a binary control with a constant heat quantum) — no re-identification needed.

5 Buffer-tank model

5.1 Heat inputs (measured, no HP flow needed)

Both heat pumps charge the *same* tank. Their heat inputs are taken from the measured electrical powers, so the tank model contains no heat-pump flow:

$$Q_{\text{hp}} = \alpha P_{\text{el}} \quad \text{(ASHP)}, \qquad Q_{\text{gshp}} = \alpha_g P_{\text{el},g} \quad \text{(GSHP, §4)}, \tag{10}$$

with efficiency-proxy constants α, α_g (identified). Both inject into the top node.

5.2 Two-node stratified model (v1)

The tank is discretised into a top node (T_{top}) and a bottom node (T_{bot}) of fixed volumes. The district acts as a *heat sink*: hot water is withdrawn from the top and returns cold to the bottom, removing heat at the rate Q_{dist}. We drive the tank with this *forecasted heat* directly rather than with the draw flow $\dot{m}_d$, because the demand is forecast as a load (kW), not a flow, and reconstructing $\dot{m}_d\, c\, \Delta T$ only to collapse it back to a heat is needless — the useful product is the heat. Since extraction is from the top, the load lands on the top node, and with the district return sitting near the bottom node ($T_{\text{ret},d} \approx T_{\text{bot}}$) the whole draw $\dot{m}_d\, c(T_{\text{top}} - T_{\text{bot}}) \approx \dot{m}_d\, c(T_{\text{top}} - T_{\text{ret},d}) = Q_{\text{dist}}$ collapses onto it. A single factor β carries any constant mismatch between the forecast boundary and this tank's outlet (network/standing losses, metering point, a shared manifold); $\beta \approx 1$ when Q_{dist} is the tank-outlet heat meter. The heat pumps enter as the heat terms (10); their recirculation, conduction, and the flow-driven inter-node transport appear as one mixing coefficient κ. The node energy balances are:

$$C_{\text{top}}\, \dot{T_{\text{top}}} = Q_{\text{hp}} + Q_{\text{gshp}} - \beta\, Q_{\text{dist}} - \kappa\,(T_{\text{top}} - T_{\text{bot}}) - UA_t\,(T_{\text{top}} - T_{\text{amb}}), \tag{11}$$

$$C_{\text{bot}}\, \dot{T_{\text{bot}}} = \kappa\,(T_{\text{top}} - T_{\text{bot}}) - UA_b\,(T_{\text{bot}} - T_{\text{amb}}). \tag{12}$$

The whole tank then loses exactly $\beta\, Q_{\text{dist}}$ to the district (plus the standing losses), which is energy-consistent by construction. Here κ (W/K) is the mixing/conduction coefficient — it absorbs the fixed-speed HP recirculation and the flow-driven transport, and produces the gradual T_{top} droop of a non-ideal tank — and UA_t, UA_b are standing-loss coefficients. **Identified parameters:** $C_{\text{top}}, C_{\text{bot}}, \kappa, UA_t, UA_b, \alpha, \alpha_g, \beta$; the tank capacitances are anchored near their geometric values $\rho V_{\bullet} c$ (the buffer volume is known), which keeps β from trading off against C_{top}. *No heat-pump flow $\dot{m}_h$ appears.*

Bottom-node anchor. Dropping the explicit cold-return term removes the bottom node's cold source, so T_{bot} can drift slightly warm via κ. If the bottom fit matters, retain the measured return as a weak pull $-\gamma\,(T_{\text{bot}} - T_{\text{ret},d})$ in (12) ($T_{\text{ret},d} =$ `TT_702`, a measured input); for the top-node dynamics that drive supply and DR it is immaterial.

Opposite-curvature diagnostic (when is two-node enough?). The fixed volumes $C_{\text{top}}, C_{\text{bot}}$ are the modelling approximation: physically the stored energy lives in a *hot volume* and a *cold volume* whose boundary (the thermocline) moves, so charge/discharge is really a change of *volume* at nearly constant temperatures. Freezing the volumes forces that volume motion to be re-expressed as a *temperature* change of fixed nodes; the coefficient κ is the artifact that absorbs it. This substitution has a characteristic signature in the discharge shape of T_{top}:

- **Two-node model:** T_{top} relaxes *exponentially* toward T_{bot} — fastest drop first, then tapering (convex, decreasing slope).
- **Real moving thermocline:** T_{top} holds a *plateau* while the front is below the sensor, then drops sharply when it arrives (concave, flat-then-steep).

These are *opposite curvatures.* Hence an observed "flat for a while, then lower" discharge is the thermocline signature and the regime the two-node model is structurally worst at. Use this as a quantitative trigger: fit the two-node model first and inspect its residuals around the deepest discharges. If the model consistently *leads* the data — T_{top} predicted to droop while the measurement is still on its plateau, then catching up after the real cliff — the knee is too sharp for two nodes and the V_{top} thermocline state (§5.3) is required. For a DR controller this residual pattern is not cosmetic: it means the model under-predicts available flexibility on the plateau and then over-predicts it past the cliff, i.e. it mistimes exactly the moment the tank runs out. A gentle knee (soft plateau, mild drop) stays inside the two-node regime and κ can absorb it.

Validity. A two-node model reproduces a *smooth* T_{top} decay during discharge *by construction*: it cannot represent a sharp thermocline sweeping past the sensor. It is therefore valid only in the regime where the real tank actually de-stratifies smoothly. This should be checked on the deepest observed discharges; where T_{top} instead drops abruptly (a "cliff"), use the thermocline state below.

5.3 Thermocline state (v1.5, optional)

The point sensors $T_{\text{top}}, T_{\text{bot}}$ sit near the tank ends. Their value is not the usable stored energy: a thin hot layer over a large cold volume stores little. Introduce the *hot-zone volume* V_{top} (position of the thermocline). Because the HP loop is not flow-metered, its *volumetric* charging rate is recovered from the measured *electrical* power via the heat balance:

$$\dot{V}_{\text{charge}} = \frac{Q_{\text{hp}}}{\rho\, c\,(T_{\text{top}} - T_{\text{bot}})} = \frac{\alpha\, P_{\text{el}}}{\rho\, c\,(T_{\text{top}} - T_{\text{bot}})}, \qquad Q_{\text{hp}} \approx \alpha\, P_{\text{el}}, \tag{13}$$

while the discharge draws hot water from the top at the measured district flow times the fraction pulled from the tank by the valve, $g(\theta)$:

$$\boxed{\dot{V_{\text{top}}} = \frac{\alpha\, P_{\text{el}}}{\rho\, c\,(T_{\text{top}} - T_{\text{bot}})} - \dot{m}_d\, g(\theta)} \tag{14}$$

Equation (14) degenerates as $T_{\text{top}} - T_{\text{bot}} \to 0$ (no distinct thermocline); clamp/freeze V_{top} below a ΔT threshold.

Observability of V_{top} from point sensors. V_{top} is *not* continuously measured, but it is partially observed as a *threshold event*: while the thermocline is below the top sensor, T_{top} saturates at the hot value ($V_{\text{top}} \geq V_{\text{sensor}}$, exact value unknown); the instant T_{top} drops is a hard fix that the front has reached the sensor height. This yields a predict/correct observer:

$$\begin{aligned} &\text{predict:} \quad \dot{V_{\text{top}}} \text{ by (14) (open-loop integration)}, && (15)\\ &\text{correct:} \quad V_{\text{top}} \leftarrow V_{\text{sensor}} \text{ when a } T_{\text{top}} \text{ (or } T_{\text{bot}}\text{) crossing is detected.} && (16) \end{aligned}$$

Between the two end-sensors the state is unobserved and relies on the integration; the crossings cancel its drift.

6 Distribution valve and heating curve

The 3-way valve blends tank-top water with the district return to deliver the supply temperature:

$$T_{s,d} = \theta\, T_{\text{top}} + (1-\theta)\, T_{\text{ret},d}, \qquad \theta \in [0,1]. \tag{17}$$

A second internal PID drives θ so that $T_{s,d} \to T^\star_{s,d}$. **We do not model this PID and do not identify its gains**: the valve is not a demand-response actuator (it is absent from the MPC decision variables in (30)), so its dynamics are irrelevant to the flexibility problem. The *only* thing this loop contributes to our model is its steady-state result — the algebraic feasibility constraint (19) — because whenever the tank is hot enough the valve holds $T_{s,d} = T^\star_{s,d}$ exactly, and when it is not the valve saturates. That is all the MPC needs from the valve.

The set-point $T^\star_{s,d}$ is the weather-compensated heating curve (encoded in the winter scenario, columns `MODE_HIV_ConsigneSupplyIfTempExt*Param`); a two-level day/night form is

$$T^\star_{s,d}(T_{\text{air}}) = \begin{cases} T_{\text{cold}}, & T_{\text{air}} < \bar{T} - \frac{\Delta}{2}, \\ T_{\text{mild}}, & T_{\text{air}} > \bar{T} + \frac{\Delta}{2}, \end{cases} \tag{18}$$

with switch threshold $\bar{T}$ (`MODE_HIV_paramTemperatureExt`) and hysteresis Δ (`paramOffsetTempExt`).

Feasibility (the key coupling). Blending (17) can only *raise* the return toward T_{top}; the valve saturates at $\theta = 1$ once $T_{\text{top}} = T^\star_{s,d}$. Hence the deliverable-supply constraint

$$T_{s,d} = T^\star_{s,d} \iff T_{\text{top}} \geq T^\star_{s,d}(T_{\text{air}}). \tag{19}$$

The right-hand inequality is the *floor* of the usable storage: once T_{top} falls to $T^\star_{s,d}$, flexibility is exhausted and the heat pumps must recharge.

7 Reduced model and MPC formulation

7.1 Collapsing the PID loops

The low-level PIDs (HP loop (3) and the valve loop of §6) are not re-optimised by the MPC. On the MPC time step they are assumed to reach their set-points whenever physically feasible, which replaces the fast dynamics by algebraic relations plus saturation limits:

$$T_{hs} = T_{\mathrm{sp}} \qquad \text{while } m < 1 \;\; (Q_{\mathrm{hp}} < \dot{Q}_{\max}(T_{\mathrm{air}})), \tag{20}$$

$$T_{s,d} = T^{\star}_{s,d}(T_{\mathrm{air}}) \qquad \text{while } T_{\mathrm{top}} \geq T^{\star}_{s,d}(T_{\mathrm{air}}). \tag{21}$$

The heat-pump heat input, expressed directly from the decision variable (needed because in a forward prediction P_{el} is an output, not yet known):

$$Q_{\mathrm{hp}} = \min\Big(G_h\,(T_{\mathrm{sp}} - T_{\mathrm{bot}}),\;\; \dot{Q}_{\max}(T_{\mathrm{air}}) \Big), \qquad G_h \equiv \dot{m}_h\, c, \tag{22}$$

where G_h (W/K) is the *single lumped condenser conductance* of §5: the unknown fixed pump flow times c, identified as one constant. The two limbs are the compressor working ($T_{hs} = T_{\mathrm{sp}}$) and the compressor saturated at full modulation.

7.2 Reduced dynamics

Dropping the fast condenser states (7)–(8), the control model keeps the two tank states (optionally V_{top}):

$$C_{\mathrm{top}}\,\dot{T}_{\mathrm{top}} = Q_{\mathrm{hp}} + Q_{\mathrm{gshp}} - \beta\, Q_{\mathrm{dist}} - \kappa(T_{\mathrm{top}} - T_{\mathrm{bot}}) - UA_t(T_{\mathrm{top}} - T_{\mathrm{amb}}), \tag{23}$$

$$C_{\mathrm{bot}}\,\dot{T}_{\mathrm{bot}} = \kappa(T_{\mathrm{top}} - T_{\mathrm{bot}}) - UA_b(T_{\mathrm{bot}} - T_{\mathrm{amb}}), \tag{24}$$

$$P_{\mathrm{el}} = Q_{\mathrm{hp}} \,/\, \mathrm{COP}(T_{\mathrm{air}}, T_{\mathrm{sp}}). \tag{25}$$

7.3 Constraints (where the PID logic becomes explicit)

$$T_{\mathrm{top}} \geq T^{\star}_{s,d}(T_{\mathrm{air}}) \qquad \text{valve saturation(19) : DR floor,} \tag{26}$$

$$T_{\mathrm{top}} \leq T_{\max} \qquad \text{tank ceiling,} \tag{27}$$

$$Q_{\mathrm{hp}} \leq \dot{Q}_{\max}(T_{\mathrm{air}}) \qquad \text{compressor saturation } (m = 1), \tag{28}$$

$$T_{\mathrm{sp}} \in [T_{\mathrm{sp}}^{\min}, T_{\mathrm{sp}}^{\max}], \quad \delta \in \{0, 1\} \qquad \text{actuator limits.} \tag{29}$$

7.4 Objective

Minimise energy cost net of PV self-consumption over the horizon H:

$$\min_{\{T_{\mathrm{sp}}(t),\,\delta(t)\}} \int_0^H \Big[\; \lambda(t)\, P_{\mathrm{el}}(t) \;-\; c_{\mathrm{pv}}\, \min\big(P_{\mathrm{el}}(t), \mathrm{PV}(t)\big) \;\Big]\, \mathrm{d}t. \tag{30}$$

7.5 Convexity note

The plant is nonconvex (bilinear tank advection; $P_{\mathrm{el}} = Q_{\mathrm{hp}}/\mathrm{COP}$). Two routes:

1. **Nonlinear MPC:** solve (23)–(30) directly as an NLP (e.g. CasADi + IPOPT). Recommended for the first closed loop.
2. **Convex approximation:** freeze the COP on the *forecast* exogenous signals, $\mathrm{COP}(t) = \mathrm{COP}(T_{\mathrm{air}}(t), T^{\star}_{s,d}(t))$, so the cost $\lambda(t)/\mathrm{COP}(t) \cdot Q_{\mathrm{hp}}$ becomes linear in Q_{hp}; with a fixed-node tank (23)–(24) the problem is an LP/QP. This ignores the efficiency penalty of charging hotter, to be reintroduced by piecewise-linear COP later.

8 System identification with `ctsmTMB`

Identification is carried out with `ctsmTMB` (continuous–discrete stochastic state-space models; maximum likelihood via an extended/unscented Kalman filter). The model is a set of Itô SDEs for the states plus discrete observation equations for the sensors:

$$\mathrm{d}x = f(x, u, t; \theta)\,\mathrm{d}t + \sigma(\theta)\,\mathrm{d}\omega, \qquad \text{(system, process noise } \omega) \tag{31}$$

$$y_k = h(x_k, u_k; \theta) + e_k, \quad e_k \sim \mathcal{N}(0, \Sigma_e), \qquad \text{(observation, measurement noise)} \tag{32}$$

and θ is estimated by maximising the Kalman-filter likelihood. The process noise term is the reason to prefer this over plain least squares: it lets the filter absorb the *structural* error of the model (e.g. the thermocline approximation of §5) instead of biasing the parameters into it.

8.1 The reframing: the heat pumps enter as measured inputs, not maps

The apparent obstacle is that the "HP static map" (4)–(6) is nonlinear, and Q_{hp} is unmeasured. Both dissolve once one observes that **during identification the electrical power is logged, not predicted**. We never turn a set-point into power here; the machine already did that, and the result is the measured column P_{el}. Hence the heat pumps enter the tank SDE as *measured inputs* scaled by one constant each:

$$Q_{\text{hp}}(t) = \alpha \underbrace{P_{\text{el}}(t)}_{\text{input}}, \qquad Q_{\text{gshp}}(t) = \alpha_g \underbrace{P_{\text{el},g}(t)}_{\text{input}}. \tag{33}$$

There is no $\dot{Q}_{\max}$, no sat, no COP curve in the identification model: all of that is already baked into the measured P_{el}. The parameters α, α_g *are* the effective COPs, and the tank temperatures respond in proportion to the injected heat, so they are identifiable. *The internal PID gains K_p, K_i never appear* — the loop's output (P_{el}, T_{hs}) is measured.

8.2 Closed-loop caveat: the ASHP enable signal is generated by T_{top}

Section 8.1 treats P_{el} (and, by extension, δ) as a *measured input* to be fed into the tank SDE without further comment. On the K-0001 data this needs a caveat: the existing (pre-MPC) plant logic that generated the historical δ, P_{el} record is not free/exogenous, it is a **feedback law on T_{top} itself** — a Schmitt-trigger (hysteresis) relay,

$$\delta(t) = \begin{cases} 1 & T_{\text{top}}(t) \text{ crosses below } 31\,^\circ\text{C (enable)}, \\ 0 & T_{\text{top}}(t) \text{ crosses above } 35\,^\circ\text{C (disable)}, \\ \delta(t^-) & \text{otherwise (inside the hysteresis band).} \end{cases} \tag{34}$$

Empirical evidence (11-day window, 2026-05-13 to 2026-05-24, 5 min sampling).

- Raw correlation $\text{corr}(P_{\text{el}}, T_{\text{top}}) = -0.57$ — the wrong sign for a term meant to *heat* T_{top}.
- Pre-whitened cross-correlation (AR(20) prewhitening, Box–Jenkins): peak $P_{\text{el}} \to T_{\text{top}}$ correlation is at **lag 0**, $r = -0.39$ (instantaneous, negative — not the delayed positive response direct heating would produce). By contrast $P_{\text{el}} \to T_{\text{bot}}$ peaks at lag +1 sample (+5 min), $r = +0.30$: small positive delay, correct sign, consistent with T_{bot} being a passive receiver of whatever heat is actually added.
- δ (`PAC_501_RUN`) correlates 0.95 with P_{el} (expected: same event) but *also* -0.48 with T_{top} — the enable command itself is triggered by low T_{top}, so it carries the same endogeneity as P_{el}.
- T_{sp} (`PAC_501_CONSIGNE_ACTUEL` / logged as `T_set,HP,air`) is **constant** within this window — a two-level seasonal flag over the full ∼2.7-year log, not something varied independently for identification purposes. No clean exogenous proxy is available in the historical record.

Why this matters for Stage 1. The Kalman-filter MLE in (31)–(32) *implicitly treats the input trajectory as independent of the process noise driving the state. When δ (and hence P_{el}) is actually generated by* (34) *— a function of T_{top}, the very state being estimated — that independence fails, and the estimator can no longer tell "the plant responds this way to heat" apart from "the old controller reacts this way to temperature." A preliminary two-node fit on this data (constant $\bar{Q}_{\text{gshp}}$, δ and P_{el} used as raw measured inputs exactly as in* (33)*) showed the signature of this: C_{top}, C_{bot}, κ and α all pinning at whatever bound they were given rather than converging, and the two channels' noise parameters (σ_{y_1} vs. σ_{y_2}) splitting to opposite bounds — one channel's noise absorbing the mismatch instead of a real fit.*

Direct closed-loop identification (fitting the plant on closed-loop data, input treated as given) is not automatically wrong — it is provably consistent under prediction-error methods provided *the model structure is correct and the noise is correctly specified (Ljung,* System Identification: Theory for the User*). The practical reading here: closed-loop data is much less forgiving of structural misspecification (wrong injection node, missing dynamics, oversimplified coupling) than open-loop data, because feedback narrows the input's information content and concentrates any structural error into biased parameter estimates rather than averaging it out.*

Fix: fold the known switching law into the SDE. Because (34) is fully specified (even though the in-band modulation level is not), it should be written *into* the state equation rather than passed in as an external regressor whose relationship to T_{top} the estimator is left to (mis)discover on its own:

$$\mathrm{d}T_{\text{top}} = \tfrac{1}{C_{\text{top}}}\Big[\bar{P}_{\text{HP}}\,\delta(t) + \bar{Q}_{\text{gshp}}\delta_g - \beta Q_{\text{dist}} - \kappa(T_{\text{top}} - T_{\text{bot}}) - UA_t(T_{\text{top}} - T_{\text{amb}})\Big]\mathrm{d}t + \sigma_t\,\mathrm{d}\omega_t, \quad (35)$$

with $\delta(t)$ given by (34) (a state-dependent hysteresis switch, structurally identical to how δ_g already enters via $\bar{Q}_{\text{gshp}}\delta_g$ in §4) rather than the free-floating αP_{el} term of (33). The only unknown left in the ASHP term is $\bar{P}_{\text{HP}}$, the effective heat rate while enabled — once the *timing* ambiguity is removed by the known law, the *level* becomes a well-posed, identifiable parameter again, estimated directly from how fast T_{top} rises during each known-enabled stretch.

Caveats.

1. This only identifies the dynamics *within the operating band the thermostat has actually visited* (roughly 31 °C to 35 °C, plus the transients into/out of it). It says nothing about the tank's response to conditions the old controller never explored — which matters once the MPC starts commanding T_{sp} outside that band. A short designed identification test (step changes or a simple dithered sequence directly on T_{sp}, independent of T_{top}) remains the clean long-term fix, and is required before trusting the model much beyond 31 °C to 35 °C.
2. $\bar{P}_{\text{HP}}$ is an *average* effective rate over each enabled stretch. If the true in-band modulation depends on more than just enabled/disabled (e.g. how far below 31 °C the stretch started, or ramp-up dynamics), that is extra structure not captured by (34) alone — it only pins down *when* the pump runs, not *how hard*.

8.3 Stage 1 — tank dynamics (both heat pumps together)

Two states $x = [T_{\text{top}}, T_{\text{bot}}]$, driven by the tank balances of §5 promoted to SDEs by adding a process-noise term to each:

$$\mathrm{d}T_{\text{top}} = \tfrac{1}{C_{\text{top}}}\Big[\alpha P_{\text{el}} + \bar{Q}_{\text{gshp}}\delta_g - \beta Q_{\text{dist}} - \kappa(T_{\text{top}} - T_{\text{bot}}) - UA_t(T_{\text{top}} - T_{\text{amb}})\Big]\mathrm{d}t + \sigma_t\,\mathrm{d}\omega_t, \quad (36)$$

$$\mathrm{d}T_{\text{bot}} = \tfrac{1}{C_{\text{bot}}}\Big[\kappa(T_{\text{top}} - T_{\text{bot}}) - UA_b(T_{\text{bot}} - T_{\text{amb}})\Big]\mathrm{d}t + \sigma_b\,\mathrm{d}\omega_b. \quad (37)$$

The two tank sensors give the discrete observations

$$y_k^{(1)} = T_{\text{top}}(t_k) + e_k^{(1)}, \qquad y_k^{(2)} = T_{\text{bot}}(t_k) + e_k^{(2)}, \qquad e_k^{(j)} \sim \mathcal{N}(0, \varepsilon_j^2), \quad (38)$$

mapped to columns TT_601 and TT_602. The *measured inputs* $u = [P_{\mathrm{el}},\ \delta_g,\ Q_{\mathrm{dist}},\ T_{\mathrm{amb}}]$ are supplied as known time series (Q_{dist} being the forecasted district load, not a flow); the *estimated parameters* are

$$\theta = \{\, C_{\mathrm{top}},\, C_{\mathrm{bot}},\, \kappa,\, UA_t,\, UA_b,\, \alpha,\, \beta,\, \bar{Q}_{\mathrm{gshp}},\, \sigma_t, \sigma_b,\, \varepsilon_1, \varepsilon_2 \,\}.$$

ctsmTMB runs the Kalman filter for a candidate θ, reads off the likelihood of the observed TT_601,TT_602 series, and maximises it over θ. Every parameter is pinned by a distinct feature of the data: α by how fast T_{top} rises per unit P_{el}; β by the T_{top} droop per unit forecast Q_{dist} during discharge (with C_{top} anchored to the known volume so the two do not trade off); $UA_\bullet$ by the standing cool-down when both pumps are off; κ by how quickly T_{bot} follows T_{top}; $\bar{Q}_{\mathrm{gshp}}$ as explained next.

GSHP: one constant, fitted here in Stage 1 — there is no "Stage 3". The GSHP is **not** a separate stage and does **not** join the Stage-2 static equation (39): that equation regresses power against a *modulating* set-point, and the on/off GSHP has none. Because it is pure on/off with a nearly constant output, its whole model is a *single scalar* $\bar{Q}_{\mathrm{gshp}}$ (the heat it delivers while running), switched by the measured binary $\delta_g =$ PAC_301_RUN $\in \{0,1\}$ — the term $\bar{Q}_{\mathrm{gshp}}\delta_g$ in (36). There is nothing to curve-fit: flow$\times\Delta T$ or COP$\times$power can never be separated from on/off data, and the tank cares only about the net heat, which is the one number $\bar{Q}_{\mathrm{gshp}}$. It is identified *for free* inside the same fit: during the intervals when $\delta_g = 1$, the tank heats faster than αP_{el} alone can explain, and that unexplained extra rate of rise *is* $\bar{Q}_{\mathrm{gshp}}$. One number, pinned by one effect. All data are kept — when $\delta_g = 0$ the term simply vanishes, so no GSHP-off filtering is needed. (Only if the GSHP were ever metered would the $\alpha_g P_{\mathrm{el},g}$ form of (9) apply; for on/off it is unnecessary.)

8.4 Stage 2 — the actuator map (offline, all-measured)

The nonlinear COP/$\dot{Q}_{\max}$ map is needed *only by the controller*, to predict P_{el} (the cost) and the ceiling from a *hypothetical* decision T_{sp} — which never occurs in the data. It is therefore fit *separately and offline*, and it is easy because **every variable in it is measured**. Using $T_{hs} \approx T_{\mathrm{sp}}$ at steady state and (22), the *entire* static actuator — demand branch, capacity ceiling, and COP — collapses into a **single scalar equation**:

$$P_{\mathrm{el}} = \frac{\overbrace{\min\big(\, G_h(T_{\mathrm{sp}} - T_{hr}),\ a_0 + a_1 T_{\mathrm{air}} \,\big)}^{Q_{\mathrm{hp}}\ =\ \text{thermal demand, capped by ceiling}}}{\underbrace{b_0 + b_1 T_{\mathrm{air}} + b_2 T_{\mathrm{sp}}}_{\mathrm{COP}(T_{\mathrm{air}}, T_{\mathrm{sp}})}} \ \equiv\ g(x;\, p), \quad x = (T_{\mathrm{sp}}, T_{hr}, T_{\mathrm{air}}),\ \ p = (G_h, a_0, a_1, b_0, b_1, b_2). \tag{39}$$

The numerator is the delivered heat Q_{hp}: the thermal demand $G_h(T_{\mathrm{sp}} - T_{hr})$ limited by the air-dependent ceiling $a_0 + a_1 T_{\mathrm{air}}$; dividing by the COP turns heat into electrical power. All six parameters p are fit from this one equation. The GSHP is deliberately *not* in it — being on/off it carries no set-point to regress against, so it degenerates to one constant and is folded into Stage 1 (above), not fitted here.

How a nonlinear static fit works. Each logged sample i (a timestamp with the ASHP running) supplies measured regressors $x_i = (T_{\mathrm{sp}}, T_{hr}, T_{\mathrm{air}})_i$ and a measured output $y_i = P_{\mathrm{el},i}$. "Static" means the prediction $\hat{y}_i = g(x_i; p)$ is obtained by *plugging the row's numbers into* (39) — no time integration, no dependence on any other sample; each row is one independent point on a surface. Fitting chooses the parameters p that make the predictions sit closest to the measurements in the least-squares sense:

$$\min_p\ S(p) = \sum_{i=1}^{N} \big(y_i - g(x_i; p)\big)^2. \tag{40}$$

This is the *same* objective as fitting a straight line; the difference is only *how the minimum is found.* Because p enters through a product/ratio, there is no closed-form solution, so (40) is minimised **iteratively**: start from a guess p^0, linearise g about it through the sensitivities $J_{ij} = \partial g(x_i; p)/\partial p_j$, solve a *linear* least-squares problem for a downhill correction Δp, update $p \leftarrow p + \Delta p$, and repeat until S stops decreasing (Gauss–Newton / Levenberg–Marquardt). "Nonlinear least squares" is therefore just ordinary least squares reached by walking downhill; it needs a sensible starting guess and, in principle, can settle in a local minimum — that is the only cost of the nonlinearity.

The capacity ceiling is fitted from saturation — if the data reach it. There is no sensor for the ceiling $a_0 + a_1 T_{\text{air}}$; it is identified from *where the machine stops responding*, and the $\min(\cdot)$ in (39) sorts the samples into the two regimes that make this possible. *Below* the ceiling the compressor modulates: raising T_{sp} raises P_{el} and the supply reaches set-point, so these rows ride the demand branch and pin G_h and the COP b_j. *At* the ceiling the compressor is flat out: P_{el} plateaus and a *persistent* tracking error opens up ($T_{hs} < T_{\text{sp}}$ although the machine is on). Those saturated rows — "maxed out *and* still short" — pin a_0, a_1, and the fit places the ceiling at the *kink* where the point cloud bends from sloped to flat (equivalently, the upper envelope of P_{el} versus T_{air}).

Caveat — excitation. The ceiling is identifiable **only if the data actually hit it**. If the plant always kept headroom there are no saturated rows, a_0, a_1 are not identifiable, and this is not a defect: the data hold no information about a limit that was never reached. Then, for *reproducing* the data, drop the min and keep the demand branch alone; for the *MPC* — which may command into saturation — take the ceiling from the manufacturer's nameplate capacity curve $\dot{Q}_{\max}(T_{\text{air}})$ rather than the fit. Practical test: flag rows with P_{el} pinned near its maximum *and* $T_{hs} < T_{\text{sp}}$; if enough of them span a range of T_{air}, fit a_0, a_1; otherwise fix them from the datasheet and identify only G_h and the COP.

Scale degeneracy — why the two stages need each other. The Stage-2 regression has a scale ambiguity: multiplying G_h and every b_j by a common factor leaves P_{el} unchanged. It therefore fixes only the *shape* of the COP's dependence on $(T_{\text{air}}, T_{\text{sp}})$, not its absolute level. The absolute scale is pinned by Stage 1's α, because the tank thermal mass times the measured temperature rise is an *absolute* energy. Concretely:

Stage 1 (tank) $\Rightarrow$ absolute heat scale α; **Stage 2 (static regression)** $\Rightarrow$ conditions-dependence (shape)

Anchor Stage 2 by fixing b_0, or by constraining its horizon-average COP to the α obtained in Stage 1.

8.5 Purist alternative — one joint model

The two stages can be fused into a single maximum-likelihood problem. Add the logged set-point T_{sp} as a further input, and let the heat delivered be the *latent* (unmeasured) quantity computed from the decision variable, capped by the ceiling:

$$Q_{\text{hp}} = \min\big(G_h(T_{\text{sp}} - T_{\text{bot}}),\ a_0 + a_1 T_{\text{air}} \big). \tag{41}$$

This same Q_{hp} is used in *two* places at once. First it drives the tank top node, i.e. αP_{el} in (36) is replaced by Q_{hp}:

$$\mathrm{d}T_{\text{top}} = \tfrac{1}{C_{\text{top}}}\Big[Q_{\text{hp}} + \bar{Q}_{\text{gshp}}\delta_g - \beta Q_{\text{dist}} - \kappa(T_{\text{top}} - T_{\text{bot}}) - UA_t(T_{\text{top}} - T_{\text{amb}})\Big]\mathrm{d}t + \sigma_t\,\mathrm{d}\omega_t. \tag{42}$$

Second, it appears in a *third observation equation* against the measured electrical power (`PAC_501_PUISSANCE`):

$$y_k^{(3)} = \frac{Q_{\text{hp}}(t_k)}{b_0 + b_1 T_{\text{air}} + b_2 T_{\text{sp}}} + e_k^{(3)}, \qquad e_k^{(3)} \sim \mathcal{N}(0, \varepsilon_3^2). \tag{43}$$

Because the tank observation (38) and the power observation (43) now share the *same* latent Q_{hp}, the scale degeneracy of Stage 2 is resolved automatically: the tank fixes the absolute size of Q_{hp} and the power equation fixes how it splits into G_h and the COP denominator. One MLE then returns $\{C_\bullet, \kappa, UA_\bullet, \beta, \bar{Q}_{\mathrm{gshp}}, G_h, a_0, a_1, b_0, b_1, b_2, \sigma_\bullet, \varepsilon_\bullet\}$ — tank *and* actuator parameters, mutually consistent.

Trade-off. This is elegant but harder to initialise and debug: more local optima, the non-smooth $\min(\cdot)$ kink in (41), and α is now *implied* by G_h and the COP rather than fit as its own number. **Recommendation:** run the two-stage version first, then — only if the two stages disagree — move to the joint model, using the two-stage estimates as its starting guess.

8.6 Remaining steps

1. **Thermocline (optional):** calibrate the sensor-crossing heights and $g(\theta)$ for the V_{top} observer (14); trigger it by the opposite-curvature diagnostic of §5.
2. **Validation:** multi-step-ahead (k-step) prediction on held-out weeks (not one-step filtering, which hides model error), inspecting the T_{top} fit specifically around the discharge "cliffs".

Data hygiene. Time index is UTC; cumulative counters must be differenced to rates; report and handle the known 15-minute sampling gaps (`ctsmTMB` accepts non-uniform timestamps, but the gaps must be genuine, not silently interpolated); keep the inspection cache faithful (no resampling before identification).